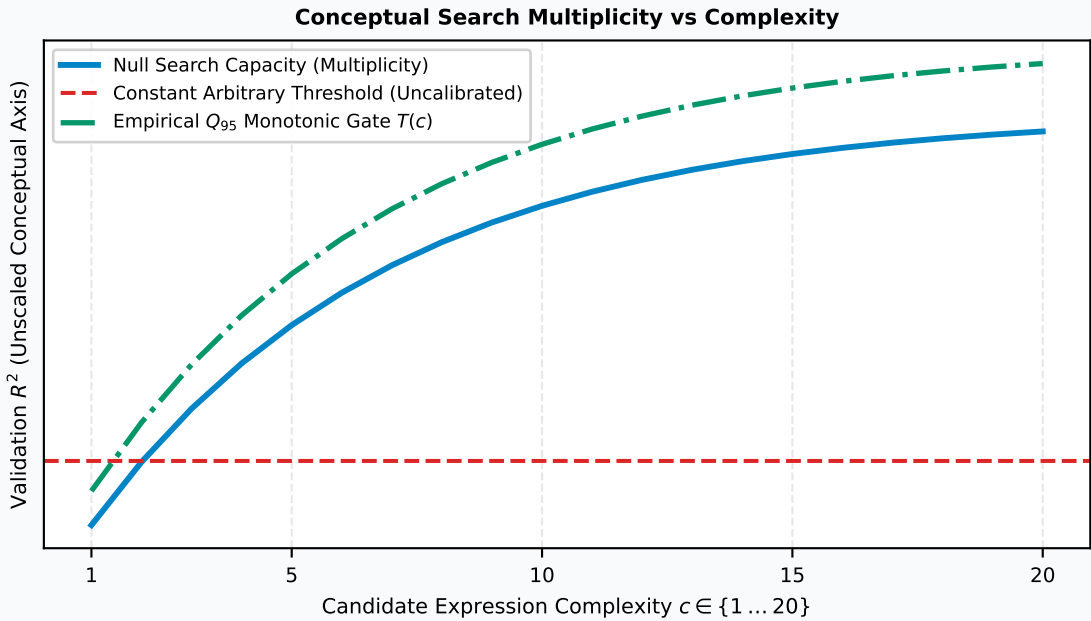


FIGURE 4: Null-Calibration Protocol, Construction Hierarchy, and Monotonic Threshold Design

A Rationale for Complexity-Indexed Empirical Thresholding

Search algorithms always return candidate expressions; threshold must match search capacity at complexity c .



B 100 Structural-Null Worlds Protocol (Amendments A3.1 & A3.2)

target_permuted_across_compounds	34 Worlds (Index 0..33)
Global compound index permutation of target values	
descriptors_permuted_across_compounds	33 Worlds (Index 34..66)
Global permutation of 5 covariate rows	
gaussian_targets_with_observed_variance	33 Worlds (Index 67..99)
Gaussian synthetic target matching observed scale	
Calibration Split: 18 / 6 / 6 Scaffolds = 108 / 36 / 36 Compounds 30 Seeds / World (3,000 Total)	

C Amendment A3.2 Base-Target Repair (Destroying Scaffold Leakage)

Rejected Provisional Design (Scaffold Confounding)

- Evaluated frozen law on unpermuted scaffolds $\rightarrow$ target inherited scaffold latent
- Permuting only covariates left target scaffold structure intact
- Produced train/val mean shifts across scaffold-disjoint splits
- Biased constant-model validation R^2 down to -0.246 (min -1.28)

Corrected A3.2 Design: Global Compound Index Permutation

- Evaluates frozen law once, then globally permutes across all 180 compounds
- Dedicated seed namespace: PB|NCAL|<world_id>|BASE_TARGET
- Preserves exact marginal distribution bitwise while destroying scaffold assignment
- Applied before split assignment (18/6/6 scaffolds) and before null transformation
- Restores clean constant-model validation $R^2 \approx -0.033$

D Threshold Construction Protocol & Calibration Shell

Mathematical Threshold Accumulation Protocol

- World Statistic:** $S(w, c) = \max_{s \in 1..30} R^2_{\text{val}}(w, s, \leq c)$
- Empirical Quantile:** $Q_{0.95}(c) = \text{numpy.quantile}(\{S(w, c)\}_{w=1}^{100}, 0.95, \text{method} = \text{'linear'})$
- Monotonic Gate:** $T(c) = \max_{1 \leq k \leq c} Q_{0.95}(k) = \text{cummax}(Q_{0.95})$
- Bootstrap Band:** 2,000 world-level resamples at seed 20260812 (reporting only)

[PROSPECTIVE RESULT PANEL — DO NOT RENDER]

Observed threshold table $T(c)$ and 2,000-resample bootstrap band remain strictly unrendered to prevent outcome visualization prior to formal unsealing. Threshold table is frozen upon calibration run completion.